

AI Model Evaluation — Questions & Answers

Reasoning, mathematics, coding, understanding, instruction-following, and difficult logic

Purpose: This PDF is designed as a simple evaluation/training reference. Answers include corrections for common reasoning and instruction-following mistakes.

Reasoning

1. Question: If all roses are flowers and some flowers fade quickly, can we conclude that some roses fade quickly?

Answer: No. We know that all roses are flowers, but we are not told that any roses are among the flowers that fade quickly.

2. Question: A farmer has 17 sheep. All but 9 run away. How many remain?

Answer: 9 sheep remain. "All but 9" means 9 stay.

3. Question: If 5 machines make 5 products in 5 minutes, how long would 100 machines take to make 100 products?

Answer: 5 minutes. Each machine makes one product in 5 minutes, so 100 machines can make 100 products in 5 minutes.

4. Question: What comes next: 2, 6, 12, 20, 30, ?

Answer: 42. The differences are 4, 6, 8, 10, so the next difference is 12.

5. Question: You have 8 balls and one is heavier. How can you find it using a balance scale in two weighings?

Answer: Weigh 3 vs 3. If one side is heavier, the heavy ball is among those 3; weigh 1 vs 1 from those 3. If the first weighing balances, the heavy ball is among the remaining 2; weigh those 2 against each other.

Math

1. Question: What is 15% of 240?

Answer: 36.

2. Question: If a \$1,000 item is discounted by 20% and then another 10%, what is the final price?

Answer: \$720. After 20%, the price is \$800; after another 10%, it is \$720.

3. Question: Solve: $3x + 7 = 28$.

Answer: $x = 7$.

4. Question: A train travels 180 km in 3 hours. What is its average speed?

Answer: 60 km/h.

5. Question: If a number is doubled and 15 is added, the result is 41. What is the number?

Answer: 13, because $2x + 15 = 41$, so $2x = 26$ and $x = 13$.

Coding

1. Question: What is a variable in programming?

Answer: A variable is a named reference used to store or represent a value that a program can use.

2. Question: What is the difference between a list and a dictionary in Python?

Answer: A list is an ordered, index-based collection. A dictionary stores key-value pairs and is accessed by keys.

3. Question: Write Python code to check whether a number is even or odd.

Answer:

```
def is_even_or_odd(n):  
    if n % 2 == 0:  
        return "even"
```

```
else:  
    return "odd"
```

4. Question: What does a for loop do?

Answer: A for loop repeatedly executes a block of code for each item in an iterable or sequence.

5. Question: Find the error in: if x = 10: print(x)

Answer: The comparison operator should be ==. Correct form: if x == 10: print(x)

Understanding

1. Question: Explain artificial intelligence to a 10-year-old.

Answer: AI is technology that lets computers perform tasks that normally require human-like abilities, such as recognizing patterns, understanding language, or making predictions.

2. Question: What is the difference between "I didn't go" and "I don't go"?

Answer: "I didn't go" refers to a specific past event that did not happen. "I don't go" usually describes a present or habitual situation.

3. Question: Summarize: "The company postponed its launch because the product required additional testing."

Answer: The company postponed its launch for additional testing.

4. Question: Rewrite "Please give me the information" in a more polite way.

Answer: Could you please provide me with the information?

5. Question: What does "break a leg" mean?

Answer: It is an idiom commonly used to wish someone good luck, especially before a performance.

Model-following

1. Question: Answer with exactly three words: What is the capital of India?

Answer: India's capital: New Delhi.

2. Question: Give me 5 fruits, but do not use bullet points.

Answer: Apples, bananas, oranges, grapes, mangoes.

3. Question: Explain photosynthesis using exactly 20 words.

Answer: Plants use sunlight, water, and carbon dioxide to produce glucose and oxygen through chlorophyll during photosynthesis, supporting life on Earth.

4. Question: Answer only YES or NO: Is 17 greater than 20?

Answer: NO.

5. Question: Give the answer first, then explain your reasoning.

Answer: Answer: 17 is less than 20. Reason: 17 is numerically smaller than 20.

More Difficult Reasoning

1. Question: A man walks 1 km south, then 1 km east, then 1 km north and ends exactly where he started. Where could he be?

Answer: The North Pole is one valid answer. There are also locations near the South Pole that can satisfy the conditions, so the North Pole is not the only possible answer.

2. Question: You have two ropes. Each takes exactly one hour to burn, but they burn at unpredictable rates. How can you measure exactly 45 minutes?

Answer: Light both ends of Rope 1 and one end of Rope 2 at the same time. Rope 1 burns out in 30 minutes. Then light the other end of Rope 2. Its remaining burn time becomes 15 minutes, giving 45 minutes total.

3. Question: There are 100 doors, initially closed. You toggle every door, then every second door, then every third door, and so on until 100. Which doors remain open?

Answer: The perfect-square doors remain open: 1, 4, 9, 16, 25, 36, 49, 64, 81, and 100. There are 10 open doors.

4. Question: A bat and ball cost ■110 together. The bat costs ■100 more than the ball. How much does the ball cost?

Answer: ■5. If the ball costs x , the bat costs $x + \text{■}100$. Then $2x + \text{■}100 = \text{■}110$, so $x = \text{■}5$.

5. Question: If yesterday was two days before Thursday, what day is today?

Answer: Wednesday. Two days before Thursday is Tuesday, so if yesterday was Tuesday, today is Wednesday.

Evaluation tip: When testing a model, check not only whether the final answer is correct, but also whether the reasoning is valid and whether exact instructions such as word counts or response formats are followed.